1. Implement Simple Naïve Bayes classification algorithm using Python on iris.csv dataset.
2. Compute Confusion matrix to find TP, FP, TN, FN, Accuracy, Error rate, Precision, Recall on the given dataset.

In [ ]:
```
# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score,
from sklearn.preprocessing import LabelEncoder, StandardScaler

pd.options.display.max_columns = None

# Set Seaborn style for visualizations
sns.set(style='whitegrid', palette='muted')
```

In [50]:
```
# Load dataset
iris = pd.read_csv('Iris.csv') if 'Iris.csv' in pd.io.common.get_handle.__glob
# Some versions include an index column; drop if present
if 'Id' in iris.columns:
    iris = iris.drop(columns=['Id'])

iris.head()
```

Out[50]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

In [51]:
```
iris.tail()
```

Out[51]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

In [52]:

```
# EDA
print('Shape:', iris.shape)
print('Size:', iris.size)
```

```
Shape: (150, 5)
Size: 750
```

In [53]:

```
print('\nInfo:')
iris.info()
```

```
Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   SepalLengthCm  150 non-null    float64
 1   SepalWidthCm   150 non-null    float64
 2   PetalLengthCm  150 non-null    float64
 3   PetalWidthCm   150 non-null    float64
 4   Species        150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

In [54]:

```
print('\nDescribe:')
iris.describe()
```

```
Describe:
```

Out[54]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

In [55]:
```
print('\nClass distribution:')
iris['Species'].value_counts()
```

```
Class distribution:
```

Out[55]:
```
Species
Iris-setosa        50
Iris-versicolor    50
Iris-virginica     50
Name: count, dtype: int64
```

In [56]:
```
# Insert rows with null values in between the dataset
null_rows = pd.DataFrame([
    {'SepalLengthCm': 6.0, 'SepalWidthCm': 3.2, 'PetalLengthCm': np.nan, 'Peta
    {'SepalLengthCm': np.nan, 'SepalWidthCm': 2.9, 'PetalLengthCm': 4.5, 'Peta
    {'SepalLengthCm': 14.0, 'SepalWidthCm': 2.9, 'PetalLengthCm': 4.5, 'Petalw
])

insert_position = 40
iris_with_null = pd.concat([iris.iloc[:insert_position], null_rows, iris.iloc[

print('New shape with inserted rows:', iris_with_null.shape)
iris_with_null.iloc[35:47]
```

```
New shape with inserted rows: (153, 5)
```

Out[56]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
35	5.0	3.2	1.2	0.2	Iris-setosa
36	5.5	3.5	1.3	0.2	Iris-setosa
37	4.9	3.1	1.5	0.1	Iris-setosa
38	4.4	3.0	1.3	0.2	Iris-setosa
39	5.1	3.4	1.5	0.2	Iris-setosa
40	6.0	3.2	NaN	2.0	Iris-virginica
41	NaN	2.9	4.5	NaN	Iris-versicolor
42	14.0	2.9	4.5	NaN	NaN
43	5.0	3.5	1.3	0.3	Iris-setosa
44	4.5	2.3	1.3	0.3	Iris-setosa
45	4.4	3.2	1.3	0.2	Iris-setosa
46	5.0	3.5	1.6	0.6	Iris-setosa

In [57]:

```
print('\nNull counts before removal:')
print(iris_with_null.isnull().sum())
```

```
Null counts before removal:
SepalLengthCm    1
SepalWidthCm     0
PetalLengthCm    1
PetalWidthCm     2
Species          1
dtype: int64
```

In [58]:

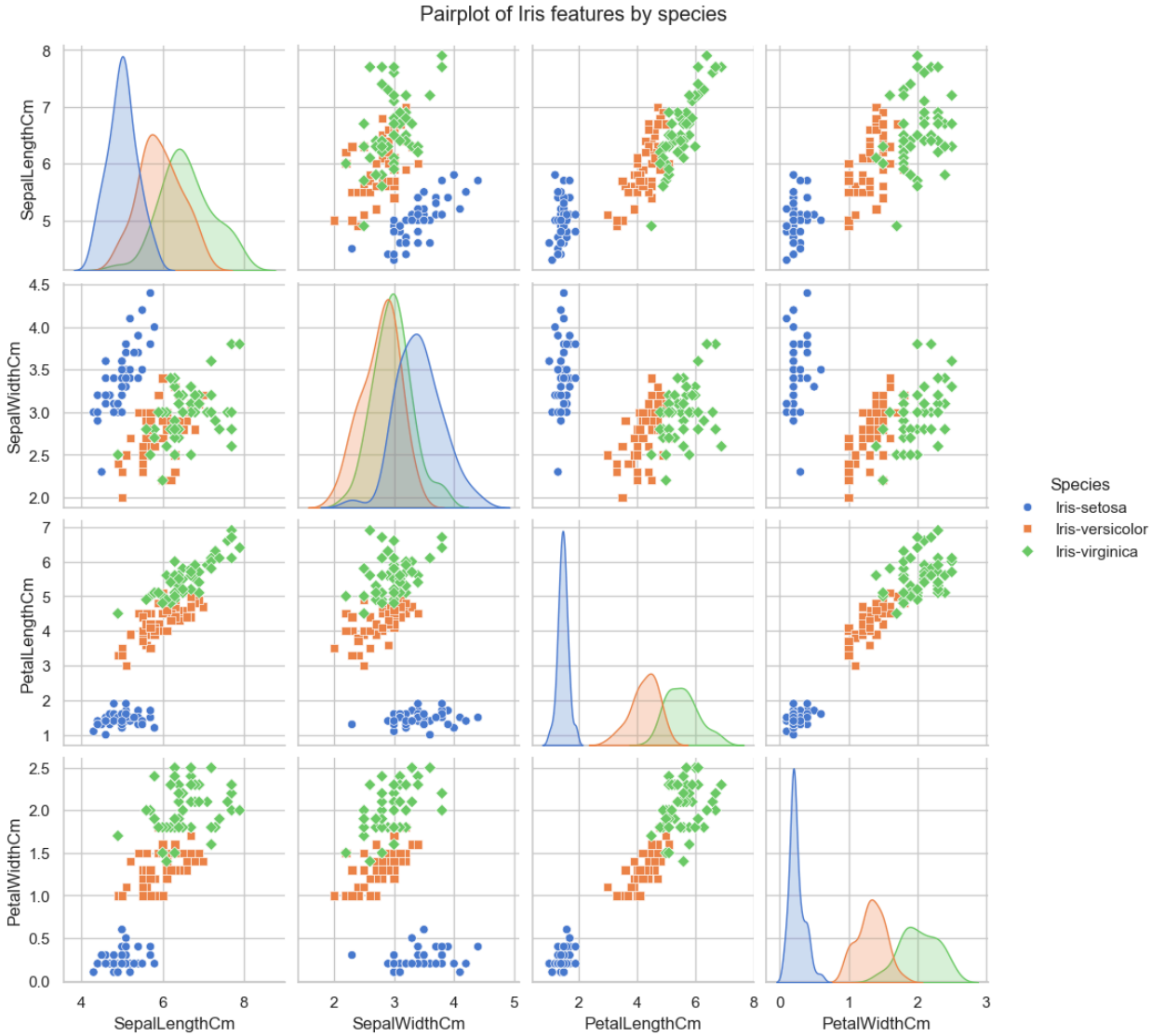
```
# Remove null values and confirm
iris_no_null = iris_with_null.dropna().reset_index(drop=True)
print('Shape after dropna:', iris_no_null.shape)
print('\nNull counts after dropna:\n', iris_no_null.isnull().sum(), sep="")
```

```
Shape after dropna: (150, 5)

Null counts after dropna:
SepalLengthCm    0
SepalWidthCm     0
PetalLengthCm    0
PetalWidthCm     0
Species          0
dtype: int64
```

In [ ]:
```
# Visualizations
plt.figure(figsize=(8, 6))
sns.pairplot(iris_no_null, hue='Species', markers=['o', 's', 'D'])
plt.suptitle('Pairplot of Iris features by species', y=1.02)
plt.show()
```

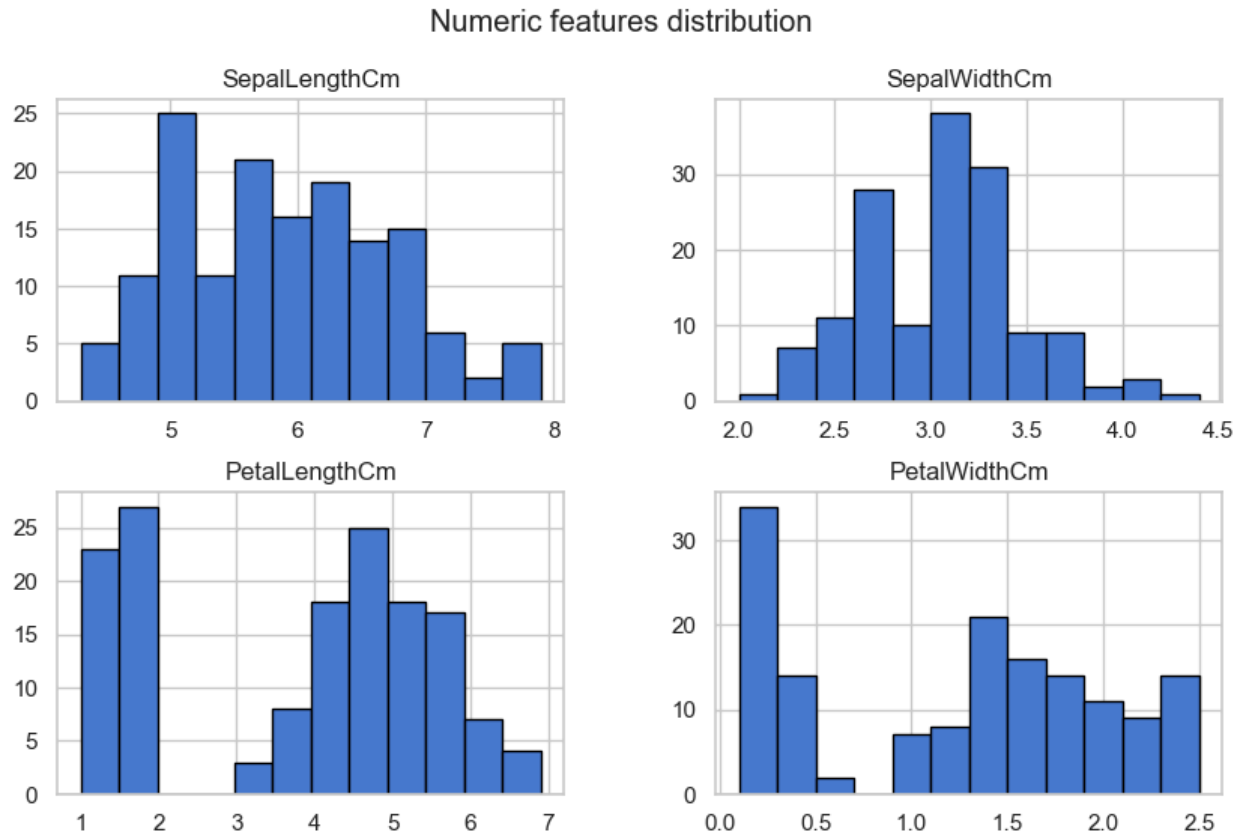
```
<Figure size 800x600 with 0 Axes>
```



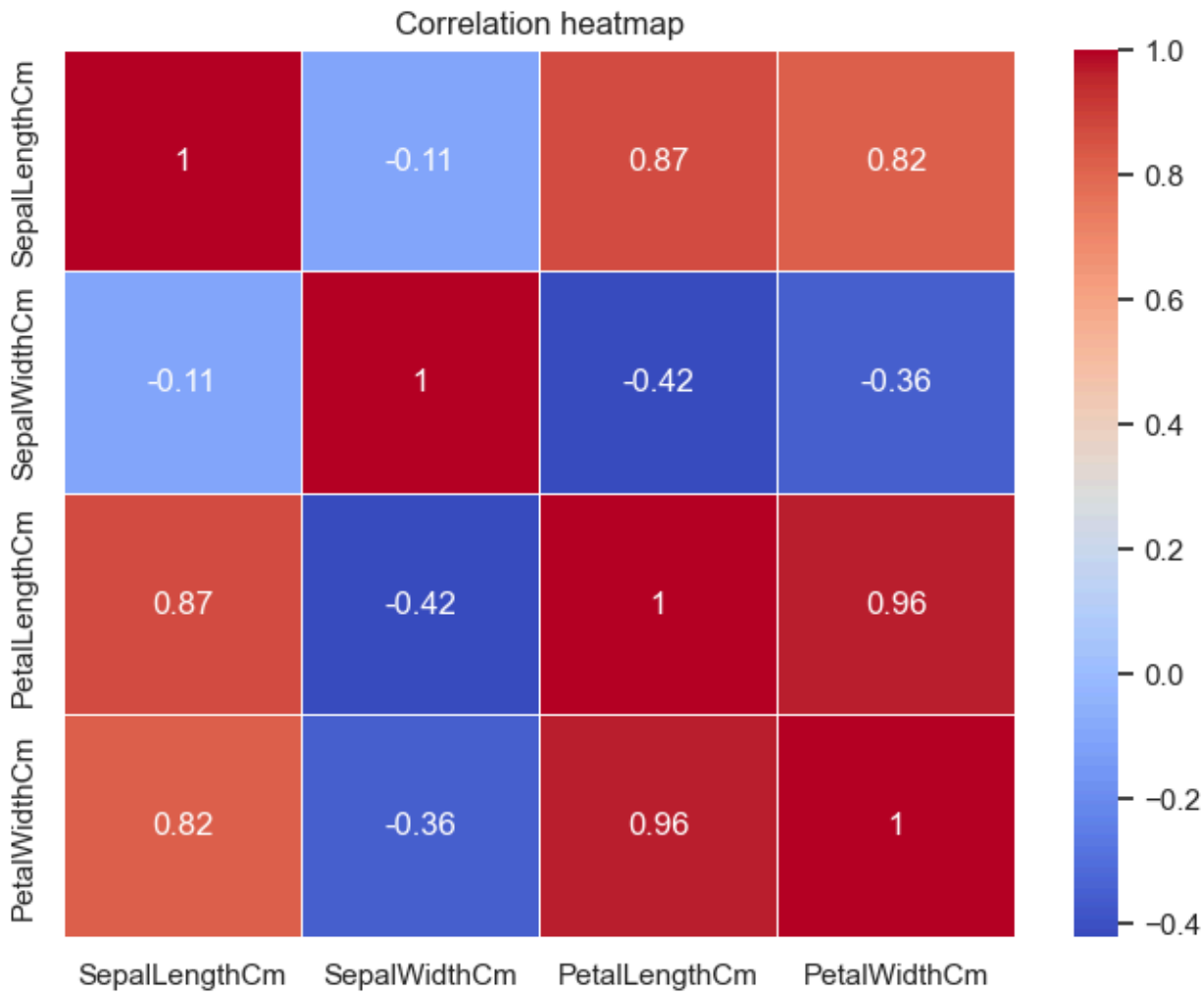
In [ ]:
```
# Distribution
plt.figure(figsize=(10, 6))
numeric_cols = ['SepalLengthCm','SepalWidthCm','PetalLengthCm','PetalWidthCm']
```

```
iris_no_null[numeric_cols].hist(bins=12, layout=(2,2), figsize=(10,6), edgecol
plt.suptitle('Numeric features distribution')
plt.show()
```

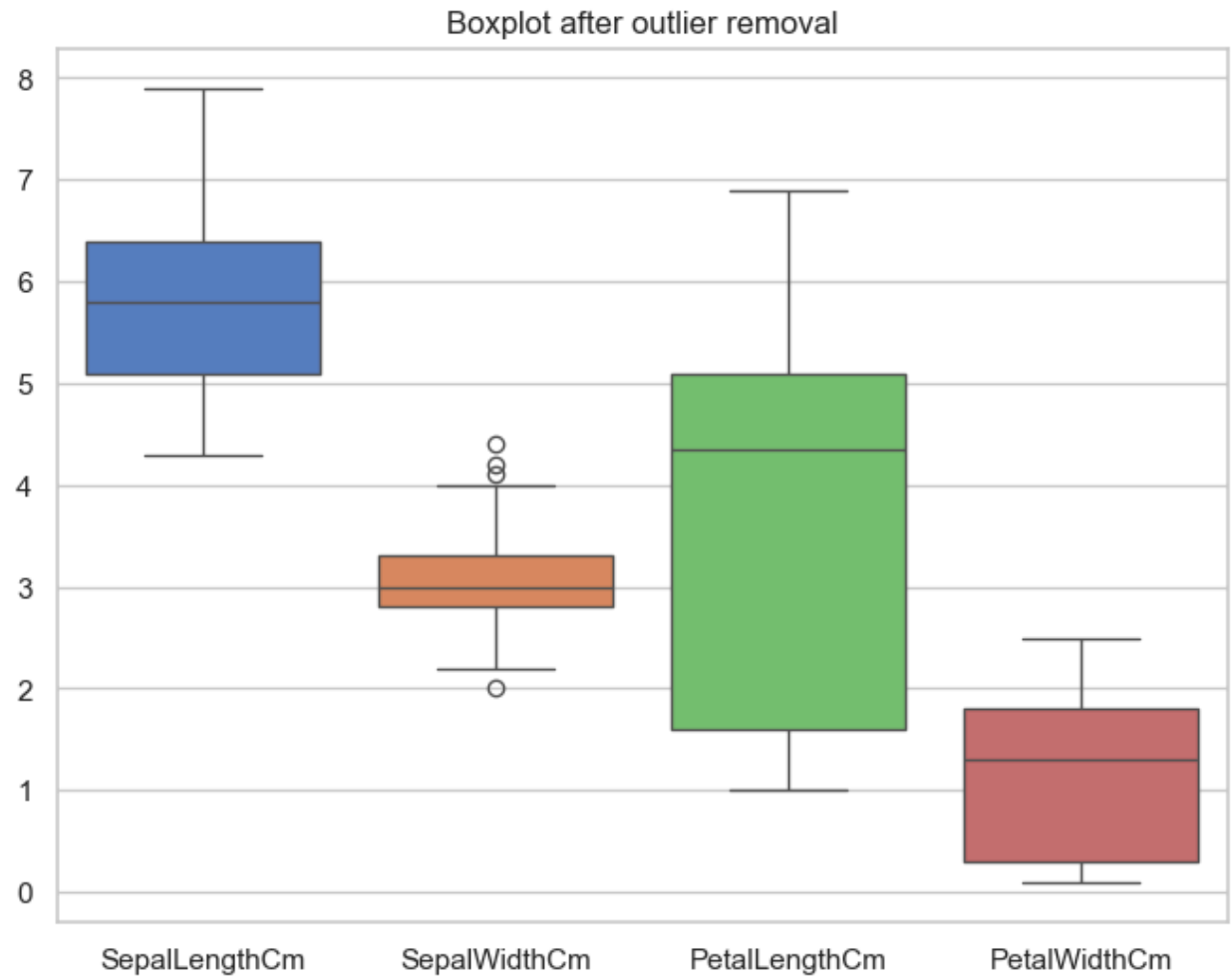
```
<Figure size 1000x600 with 0 Axes>
```



In [ ]:
```
# Correlation between features
plt.figure(figsize=(8, 6))
sns.heatmap(iris_no_null[numeric_cols].corr(), annot=True, cmap='coolwarm', li
plt.title('Correlation heatmap')
plt.show()
```



In [62]:
```
# Visualizations: boxplot
plt.figure(figsize=(8,6))
sns.boxplot(data=iris_no_null[numeric_cols])
plt.title('Boxplot after outlier removal')
plt.show()
```



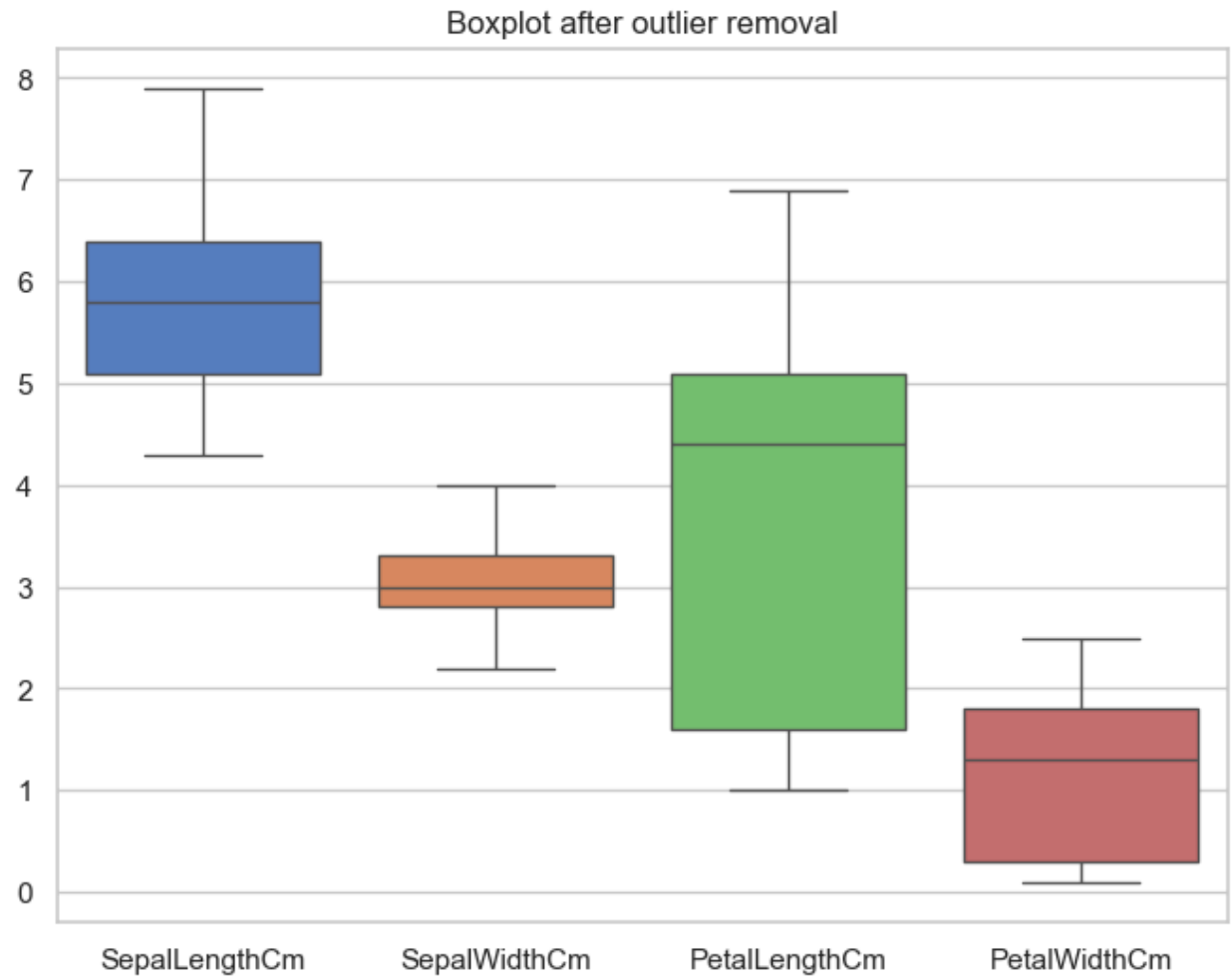
In [63]:
```
# Outlier detection and removal using IQR
iris_clean = iris_no_null.copy()

for col in numeric_cols:
    q1 = iris_clean[col].quantile(0.25)
    q3 = iris_clean[col].quantile(0.75)
    iqr = q3 - q1
    lower = q1 - 1.5 * iqr
    upper = q3 + 1.5 * iqr
    iris_clean = iris_clean[(iris_clean[col] >= lower) & (iris_clean[col] <= u

print('Shape after outlier cleanup:', iris_clean.shape)

plt.figure(figsize=(8,6))
sns.boxplot(data=iris_clean[numeric_cols])
plt.title('Boxplot after outlier removal')
plt.show()
```

```
Shape after outlier cleanup: (146, 5)
```



In [64]:
```
# Converting Categorical Column 'Species' into numeric column.
le = LabelEncoder()
iris_clean['Species_num'] = le.fit_transform(iris_clean['Species'])
print('Species mapping: ', dict(zip(le.classes_, le.transform(le.classes_))))
```

```
Species mapping:  {'Iris-setosa': np.int64(0), 'Iris-versicolor': np.int64(1),
'Iris-virginica': np.int64(2)}
```

In [65]:
```
iris_clean.head()
```

Out[65]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species	Spe
0	5.1	3.5	1.4	0.2	Iris-setosa	
1	4.9	3.0	1.4	0.2	Iris-setosa	
2	4.7	3.2	1.3	0.2	Iris-setosa	
3	4.6	3.1	1.5	0.2	Iris-setosa	
4	5.0	3.6	1.4	0.2	Iris-setosa	

In [66]:

```
# Separating Features and Labels for model.
X = iris_clean[numeric_cols].copy()
y = iris_clean['Species_num']
```

In [67]:

```
# Normalising the Data using Standard Scaler
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

print('First 5 scaled rows:')
pd.DataFrame(X_scaled, columns=numeric_cols).head()
```

```
First 5 scaled rows:
```

Out[67]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
0	-0.910515	1.175789	-1.374878	-1.345899
1	-1.151122	-0.093924	-1.374878	-1.345899
2	-1.391729	0.413961	-1.431986	-1.345899
3	-1.512032	0.160019	-1.317771	-1.345899
4	-1.030819	1.429732	-1.374878	-1.345899

In [68]:

```
# Splitting the data inot Train-Test splits
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2
print('Train shape:', X_train.shape, 'Test shape:', X_test.shape)
```

```
Train shape: (116, 4) Test shape: (30, 4)
```

In [69]:

```
# Naive Bayes classification model
nb = GaussianNB()
```

In [70]:

```
# Training model on data
nb.fit(X_train, y_train)
```

Out[70]:

GaussianNB

Parameters

priors	None
var_smoothing	1e-09

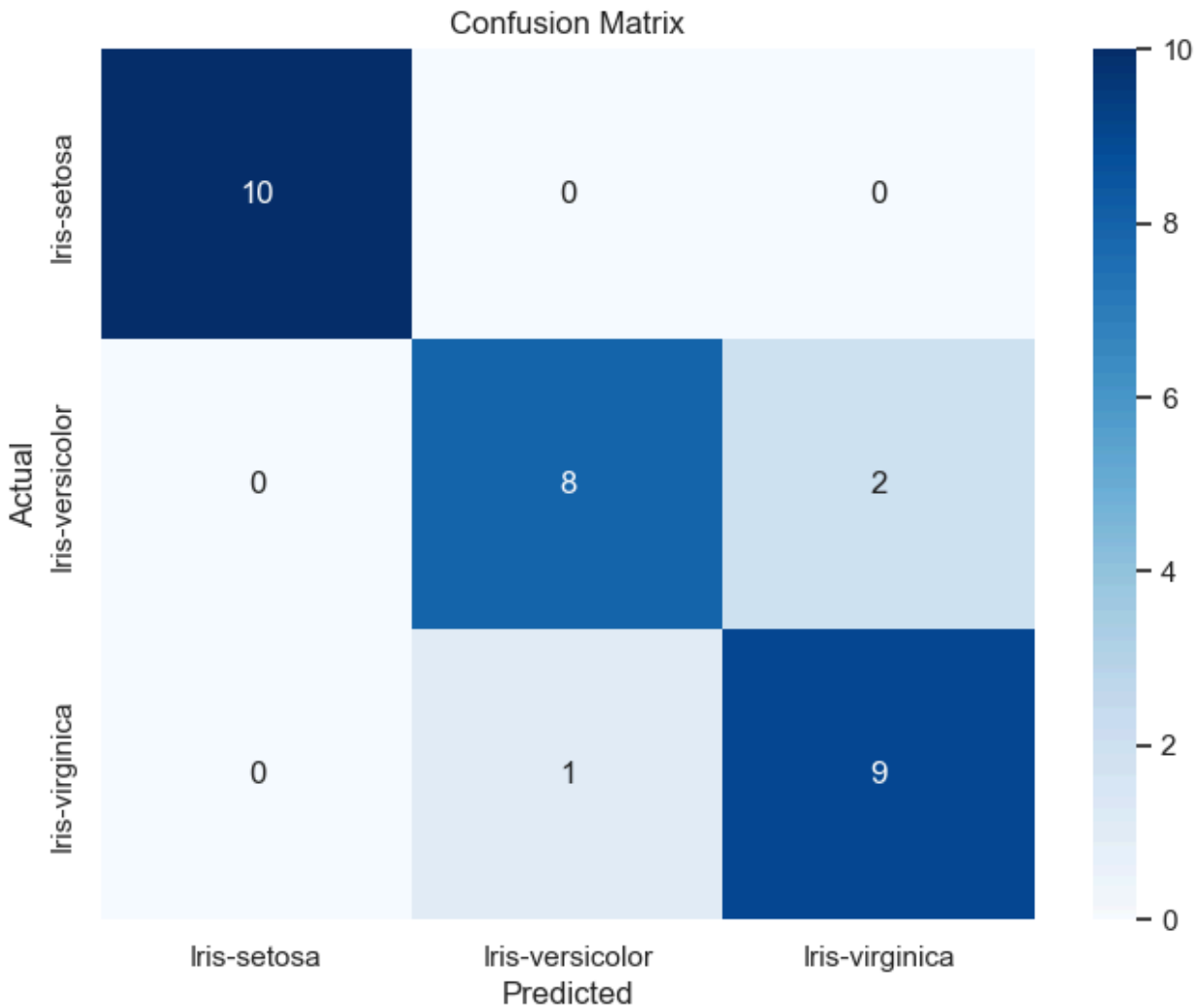
In [71]:

```
# Predicting the results on test data.
y_pred = nb.predict(X_test)
print('Predicted:', y_pred[:10])
print('Actual   :', y_test.values[:10])
```

```
Predicted: [1 1 2 0 2 1 2 0 2 2]
Actual   : [1 1 2 0 2 1 1 0 2 2]
```

In [75]:

```
# Confusion matrix
cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(8, 6))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=le.classes_, yt
plt.title('Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```



In [73]:

```
# Evaluation metrics from Confusion Matrix
mcm = multilabel_confusion_matrix(y_test, y_pred)
for idx, label in enumerate(le.classes_):
    tn, fp, fn, tp = mcm[idx].ravel()
    print(f'Class {label}: TP={tp}, FP={fp}, TN={tn}, FN={fn}')

accuracy = accuracy_score(y_test, y_pred)
error_rate = 1 - accuracy
precision = precision_score(y_test, y_pred, average='macro')
recall = recall_score(y_test, y_pred, average='macro')
f1 = f1_score(y_test, y_pred, average='macro')

print(f'Accuracy: {accuracy:.4f}')
print(f'Error rate: {error_rate:.4f}')
print(f'Precision (macro): {precision:.4f}')
print(f'Recall (macro): {recall:.4f}')
print(f'F1 score (macro): {f1:.4f}')
```

```
Class Iris-setosa: TP=10, FP=0, TN=20, FN=0
Class Iris-versicolor: TP=8, FP=1, TN=19, FN=2
Class Iris-virginica: TP=9, FP=2, TN=18, FN=1
Accuracy: 0.9000
Error rate: 0.1000
Precision (macro): 0.9024
Recall (macro): 0.9000
F1 score (macro): 0.8997
```

In [74]:

```python
# Detailed classification report
from sklearn.metrics import classification_report
print('Classification Report:')
print(classification_report(y_test, y_pred, target_names=le.classes_))
```

```
Classification Report:
                 precision    recall  f1-score   support

    Iris-setosa       1.00      1.00      1.00        10
Iris-versicolor       0.89      0.80      0.84        10
 Iris-virginica       0.82      0.90      0.86        10

       accuracy                           0.90        30
      macro avg       0.90      0.90      0.90        30
   weighted avg       0.90      0.90      0.90        30
```